

✓ Exercises XP: Vector Databases and RAG

Use this guided notebook and fill each TODO before running cells.

What you'll learn

- Vector search strategies (KNN, ANN) and evaluation.
- Vector database utility (similarity search, RAG).
- Differences between vector DBs, libraries, and plugins.
- Best practices for vector store usage and performance.
- How LMs use context; embedding generation and storage.
- Querying vector stores and applying LMs for QA with retrieved context.

What you'll build

A functional RAG pipeline with FAISS and ChromaDB, plus QA over retrieved context using a Hugging Face model.

✓ 0. Setup

Run the install cell once. If your platform needs system deps (e.g., libomp for FAISS), follow instructions in comments.

```
%pip uninstall -y pydantic-core pydantic
%pip install -U "pydantic<2"
%pip install -U "faiss-cpu>=1.8.0" "chromadb==0.3.21"
%pip install -U "numpy<2" sentence-transformers transformers
```

```
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.0.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch>=2.0.0)
Requirement already satisfied: mdurl<=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0)
```

```
import os
import json
from pathlib import Path
import numpy as np
import pandas as pd
import faiss
from sentence_transformers import SentenceTransformer, InputExample
import chromadb
from chromadb.config import Settings
from transformers import AutoTokenizer, AutoModelForCausalLM, pipeline
from IPython.display import display
os.makedirs('cache', exist_ok=True)
```

✖ 🌟 Exercise 1 · Data loading and preparation

```
data_path = 'labelled_newscatcher_dataset.csv'
pdf = pd.read_csv(data_path, sep=';')

if 'id' not in pdf.columns:
    pdf['id'] = range(len(pdf))

pdf_subset = pdf.head(1000).copy()

display(pdf_subset[['id', 'title']].head())
```

	id	title	
0	0	A closer look at water-splitting's solar fuel ...	
1	1	An irresistible scent makes locusts swarm, stu...	
2	2	Artificial intelligence warning: AI will know ...	
3	3	Glaciers Could Have Sculpted Mars Valleys: Study	
4	4	Perseid meteor shower 2020: What time and how ...	

✖ 🌟 Exercise 2 · Vectorization with Sentence Transformers

```
def example_create_fn(idx: int, text: str) -> InputExample:
    return InputExample(guid=str(idx), texts=[text], label=0.0)

faiss_train_examples = [
    example_create_fn(row['id'], row['title'])
    for _, row in pdf_subset.iterrows()
]

faiss_train_examples[:2]
```

```
[<sentence_transformers.sentence_transformer.readers.input_example.InputExample at 0x781edd0b78c0>,
 <sentence_transformers.sentence_transformer.readers.input_example.InputExample at 0x781edce67a70>]
```

```
model = SentenceTransformer('all-MiniLM-L6-v2')

titles_list = pdf_subset['title'].tolist()

faiss_title_embedding = model.encode(titles_list, convert_to_numpy=True, show_progress_bar=True)

print(f"Number of vectors: {len(faiss_title_embedding)}")
print(f"Dimensions per vector: {len(faiss_title_embedding[0])}")
```

```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:93: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/t)
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limit
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a
modules.json: 100% 349/349 [00:00<00:00, 14.9kB/s]

config_sentence_transformers.json: 100% 116/116 [00:00<00:00, 5.82kB/s]

README.md: 10.5k/? [00:00<00:00, 370kB/s]

sentence_bert_config.json: 100% 53.0/53.0 [00:00<00:00, 2.70kB/s]

config.json: 100% 612/612 [00:00<00:00, 31.3kB/s]

model.safetensors: 100% 90.9M/90.9M [00:00<00:00, 209MB/s]

Loading weights: 100% 103/103 [00:00<00:00, 2947.07it/s]

tokenizer_config.json: 100% 350/350 [00:00<00:00, 27.8kB/s]

vocab.txt: 232k/? [00:00<00:00, 8.65MB/s]

tokenizer.json: 466k/? [00:00<00:00, 15.0MB/s]

special_tokens_map.json: 100% 112/112 [00:00<00:00, 10.3kB/s]

config.json: 100% 190/190 [00:00<00:00, 18.9kB/s]

Batches: 100% 32/32 [00:10<00:00, 3.21it/s]

Number of vectors: 1000
Dimensions per vector: 384

```

✦ Exercise 3 · FAISS indexing and search

```

pdf_to_index = pdf_subset
id_index = pdf_to_index['id'].to_numpy().astype(np.int64)
content_encoded_normalized = faiss_title_embedding.astype('float32')
faiss.normalize_L2(content_encoded_normalized)
index_content = faiss.IndexIDMap(faiss.IndexFlatIP(content_encoded_normalized.shape[1]))
index_content.add_with_ids(content_encoded_normalized, id_index)
index_content.ntotal

```

1000

```

def search_content(query: str, pdf_to_index: pd.DataFrame, k: int = 3):

    query_vector = model.encode([query]).astype('float32')

    faiss.normalize_L2(query_vector)

    sims, ids = index_content.search(query_vector, k)

    results = pdf_to_index[pdf_to_index['id'].isin(ids[0])].copy()

    results['similarities'] = sims[0]

    return results.sort_values(by='similarities', ascending=False)

display(search_content('animal', pdf_to_index, k=5))

```

	topic	link	domain	published_date	title	lang	id	si
99	TECHNOLOGY	https://www.gematsu.com/2020/08/ghostwire-toky...	gematsu.com	2020-08-07 16:43:13	Ghostwire: Tokyo confirms dog petting	en	99	
176	TECHNOLOGY	https://www.pushsquare.com/news/2020/08/random...	pushsquare.com	2020-08-03 16:30:00	Random: You Can Pick Up and Pet Cats in Assass...	en	176	
762	SCIENCE	https://af.reuters.com/article/worldNews/idAFK...	af.reuters.com	2020-08-13 16:54:00	'Secret' life of sharks: Study	en	762	

✧ Exercise 4 · ChromaDB collection and querying

```
chroma_client = chromadb.Client(Settings(anonymized_telemetry=False))
collection_name = 'my_news'
if any(c.name == collection_name for c in chroma_client.list_collections()):
    chroma_client.delete_collection(name=collection_name)

collection = chroma_client.create_collection(name=collection_name)

collection.add(
    embeddings=faiss_title_embedding.tolist(),
    documents=pdf_subset['title'].tolist(),
    ids=[str(i) for i in pdf_subset['id']]
)

results = collection.query(
    query_texts=["animal"],
    n_results=5
)

print(json.dumps(results, indent=2))
```

```
ERROR:chromadb.telemetry.posthog:Failed to send telemetry event client_start: capture() takes 1 positional argument
WARNING:chromadb:Using embedded DuckDB without persistence: data will be transient
WARNING:chromadb.api.models.Collection:No embedding_function provided, using default embedding function: Sentence
Loading weights: 100% 103/103 [00:00<00:00, 2531.34it/s]
ERROR:chromadb.telemetry.posthog:Failed to send telemetry event collection_add: capture() takes 1 positional argument
{
  "ids": [
    [
      "176",
      "975",
      "99",
      "928",
      "762"
    ]
  ],
  "embeddings": null,
  "documents": [
    [
      "Random: You Can Pick Up and Pet Cats in Assassin's Creed Valhalla",
      "Researchers explore social behavior of animals toward emerging infectious diseases",
      "Ghostwire: Tokyo confirms dog petting",
      "Just Let This Lizard Be a Dinosaur",
      "'Secret' life of sharks: Study reveals their surprising social networks"
    ]
  ],
  "metadatas": [
    [
      null,
      null,
      null,
      null,
      null
    ]
  ],
  "distances": [
    [
      1.216195821762085,
      1.2464324235916138,
      1.3118828535079956,
      1.3652254343032837,
      1.409005880355835
    ]
  ]
}
```

✧ Exercise 5 · Question answering with a Hugging Face model

```
from transformers import pipeline

model_id = 'google/flan-t5-small'

pipe = pipeline("text-generation", model=model_id)

question = "What's the latest news on space development?"

context_docs = results['documents'][0][:3]
context = ' '.join(context_docs)
```

```
prompt = f"Answer the question using only the context.\nContext: {context}\nQuestion: {question}\nAnswer:\n"

response = pipe(prompt, max_length=50)[0]['generated_text']

print(f"Prompt sent to model:\n{prompt}")
print("-" * 30)
print(f"Model Response: {response}")
```

```
model.safetensors: 100% 308M/308M [00:05<00:00, 40.6MB/s]
Loading weights: 100% 190/190 [00:00<00:00, 2937.74it/s]
[transformers] The tied weights mapping and config for this model specifies to tie shared.weight to lm_head.weight
generation_config.json: 100% 147/147 [00:00<00:00, 9.96kB/s]
tokenizer_config.json: 2.54k/? [00:00<00:00, 197kB/s]
spiece.model: 100% 792k/792k [00:00<00:00, 3.42MB/s]
tokenizer.json: 2.42M/? [00:00<00:00, 50.2MB/s]
special_tokens_map.json: 2.20k/? [00:00<00:00, 175kB/s]
[transformers] The model 'T5ForConditionalGeneration' is not supported for text-generation. Supported models are
[transformers] Passing `generation_config` together with generation-related arguments=({'max_length'}) is deprecated
[transformers] Both `max_new_tokens` (=256) and `max_length` (=50) seem to have been set. `max_new_tokens` will take precedence.
Prompt sent to model:
Answer the question using only the context.
Context: Random: You Can Pick Up and Pet Cats in Assassin's Creed Valhalla Researchers explore social behavior of
Question: What's the latest news on space development?
Answer:

-----
Model Response: Answer the question using only the context.
Context: Random: You Can Pick Up and Pet Cats in Assassin's Creed Valhalla Researchers explore social behavior of
Question: What's the latest news on space development?
Answer:
```